

MCA SEM-II Syllabus

Sr No	Category	Course Code	Subject	L	T	P	CA	ESE	Total	Credit
1	PCC	MC2T001	Data Base Management System	4	0	0	40	60	100	4
2	PCC	MC2T002	Artificial Intelligence	4	0	0	40	60	100	4
3	PCC	MC2T003	Data Science Using Python	3	0	0	40	60	100	3
4	PCC	MC2T004	Cyber Security & Cryptography	3	0	0	40	60	100	3
5	PEC	MC2E005	Department Elective-I	2	0	0	40	60	100	2
6	PCC	MC2L006	Data Base Management System LAB	0	0	4	60	40	100	2
7	PCC	MC2L007	Python LAB	0	0	4	60	40	100	2
8	PCC	MC2L008	PHP LAB	0	0	4	60	40	100	2
Total				16	0	12	380	420	800	22

ELECTIVE-I Basket (MC2E005)

PHP

Computer Graphics

Research Methodology

Semester	Course Code	Name of the course	L	T	P	Credits
II	MC2E005	PHP	2	-	-	2

Marks Distribution			
CA	MSE	ESE	Total
40	-	60	100

Prerequisites for the course	
1	Basic understanding of HTML and CSS.
2	Understanding of web browsers and how websites work.

Prior Reading Material/useful links	
1	PHP Manual - Official Documentation
2	W3Schools PHP Tutorial
3	MySQL Documentation

Course Objectives	
Sr No	Statement
1	To introduce the fundamentals of PHP and its role in web development.
2	To learn how to create dynamic web applications using PHP.
3	To understand the integration of PHP with databases using MySQL.
4	To develop skills in handling web forms, file management, and session management in PHP.
5	To explore advanced PHP features and apply them in developing a complete web application.

Course Outcomes		
Sr No	Code	CO statement
1	CO1	Understand and apply basic PHP syntax, variables, data types, and control structures.
2	CO2	Develop dynamic web pages by integrating PHP with HTML and managing form data.
3	CO3	Implement PHP functions, arrays, and string manipulations in web applications.
4	CO4	Integrate PHP with MySQL to perform CRUD operations and manage databases securely.
5	CO5	Develop and deploy a full-fledged web application using advanced PHP techniques and frameworks.

Unit No	Course Contents	Duration
I	Introduction to PHP and Setup - Overview of PHP: Features, Benefits, Applications - PHP Installation: XAMPP, WAMP Setup - Basic Syntax: Variables, Data Types, Constants - Control Structures: If-Else, Switch, Loops - PHP in HTML: Embedding PHP within HTML Pages	7 hours
II	PHP Functions and Working with Data - Functions: Creation, Usage, Built-in Functions - Arrays: Types, Array Functions, Sorting, Filtering - String Manipulation: Functions, Regular Expressions - Working with Dates and Times	7 hours
III	Web Forms, File Handling, and State Management - Form Handling: Methods, Validation, Security - File System Functions: Reading, Writing, Uploading Files - Session and Cookie Handling: Managing User Data - Security Measures: Input Filtering, Output Escaping	7 hours
IV	Database Connectivity and Data Management - Introduction to Databases: SQL Basics, MySQL - PHP and MySQL: Connecting, CRUD Operations - Advanced Database Concepts: Joins, Indexes, Transactions - Database Security: Preventing SQL Injection, Best Practices	8 hours
V	Advanced Topics and Project Development - Object-Oriented PHP: Classes, Objects, OOP Principles - Advanced PHP Features: Namespaces, Traits, Anonymous Classes - Building RESTful APIs with PHP - Introduction to PHP MVC Frameworks: Laravel Basics - Project Work: Develop a Complete Web Application	8 hours

Text Books				
Code	Title of the Book	Author Name /Designation / Organization	Publisher	Edition/ Publication Year
T1	PHP & MySQL: Novice to Ninja	Kevin Yank	SitePoint	6th Edition, 2017
T2	PHP for the Web: Visual QuickStart Guide	Larry Ullman	Peachpit Press	5th Edition, 2016

Reference Books				
Code	Title of the Book	Author Name/Designation/ Organization	Publisher	Edition/ Publication Year
R1	Learning PHP, MySQL & JavaScript	Robin Nixon	O'Reilly Media	5th Edition, 2018
R2	Modern PHP: New Features and Good Practices	Josh Lockhart	O'Reilly Media	1st Edition, 2015

Useful Links	
1	https://phptherightway.com/
2	https://www.w3schools.com/php
3	https://dev.mysql.com/doc/refman/8.0/en/

Semester	Course Code	Name of the course	L	T	P	Credits
II	MC2E005	Research Methodology	3	-	-	2

Marks Distribution			
CA	MSE	ESE	Total
40	-	60	100

Prerequisites for the course	
1	Familiarity with academic writing conventions and structures.
2	Basic statistical concepts and their applications in research.

Prior Reading Material/useful links	
1	https://onlinecourses.nptel.ac.in/noc23_ge36/preview
2	https://onlinecourses.nptel.ac.in/noc22_ge08/preview
3	https://www.scribbr.com/dissertation/methodology/

Course Objectives	
Sr No	Statement
1	Introduce students to the basic concepts and objectives of research and to familiarize them with the research process.
2	Enable students to distinguish between different types of research and to identify suitable research methods for various research questions.
3	Provide knowledge on sampling techniques, research design, and their applications in conducting effective and unbiased research.
4	Equip students with skills in data collection, analysis, and the use of computer applications for data processing in research.
5	Develop competencies in presenting research findings, adhering to ethical standards, and mastering various citation styles and report writing formats.

Course Outcomes		
Sr No	Code	CO statement
1	CO1	Understand the fundamentals of research, including the meaning, objectives, process, and differentiation between methods and methodology.
2	CO2	Distinguish between various types of research such as pure, applied, exploratory, descriptive, diagnostic, quantitative, and qualitative research.
3	CO3	Apply knowledge of sampling techniques and research design to develop appropriate research frameworks and reduce sampling errors.
4	CO4	Execute methods for data collection, analysis, and presentation, including primary and secondary data sources, data classification, and visualization techniques.
5	CO5	Demonstrate proficiency in research presentation, including citation styles, ethical considerations, plagiarism avoidance, and report writing.

Unit No	Course Contents	Duration
I	Fundamentals of research; Meaning, Objectives, Research process, Methods and Methodology, Criteria of good research, Review of literatures: Primary source, Secondary source, Identifying gap areas from literature review, Searching- resources, using search engines, Searching data base.	6 hours
II	Types of Research; Pure research, applied research, Exploratory Research, Descriptive research, Diagnostic research, Quantitative and Qualitative research.	7 hours
III	Research Sampling and Design: Sampling of data: Concept of sampling, Probability sampling techniques, Non probability sampling techniques, Sampling error, Research Design: Meaning, Need, Types of research design-Exploratory Research Design, components of research design and features of good research design	7 hours
IV	Methods, Collection and Analysis of Data: Types of data, Methods of data collection- Interview Method, Mailing Method, Observation Method, Survey Method etc.; Primary and secondary sources of data, Sampling- meaning and methods, Classification and Tabulation, Graphical presentation, Application of computer in research data analysis.	8 hours
V	Presentation of Research: Citation Styles- APA, MLA etc., Research ethics and Plagiarism, Indexing of journal and research output, Report writing steps in report writing, layout of report writing, reference and bibliography.	8 hours

Text Books				
Code	Title of the Book	Author Name /Designation / Organization	Publisher	Edition/ Publication Year
T1	Business Research Methods	Donald R. Cooper, Pamela S. Schindler	McGraw-Hill Education	2023
T2	Research Design: Qualitative, Quantitative, and Mixed Methods Approaches	John W. Creswell	SAGE Publications	2023

Reference Books				
Code	Title of the Book	Author Name/Designation/ Organization	Publisher	Edition/ Publication Year
R1	Research Methodology: A	Ranjit Kumar	SAGE Publications	2021

	Step-by-Step Guide for Beginners			
R2	Research Methods in Education	Louis Cohen, Lawrence Manion, and Keith Morrison	Routledge	2022

Useful Links	
1	https://www.scribbr.com/dissertation/methodology/
2	https://onlinecourses.nptel.ac.in/noc23_ge36/preview

Semester	Course Code	Name of the course	L	T	P	Credits
II	MC2E005	Computer Graphics	3	-	-	2

Marks Distribution			
CA	MSE	ESE	Total
40	-	60	100

Prerequisites for the course	
1	Basic knowledge of programming in C/C++.
2	Familiarity with fundamental mathematics concepts including linear algebra and geometry.

Prior Reading Material/useful links	
1	http://www.tutorialspoint.com/computer_graphics/computer_graphics_quick_guide.htm
2	http://www.javatpoint.com/computer-graphics-tutorial
3	http://www.coursera.org/articles/computer-graphics

Course Objectives	
Sr No	Statement
1	Introduce students to the foundational concepts of computer graphics, including hardware and software components.
2	Equip students with the skills to implement basic raster graphics algorithms for drawing and filling geometric shapes.
3	Familiarize students with the OpenGL programming environment for creating and manipulating 2D and 3D graphics.
4	Teach students how to perform transformations and viewing operations on graphical objects, both in 2D and 3D spaces.
5	Provide students with an understanding of advanced graphics topics such as clipping, hidden surface removal, and animation techniques to achieve realism in computer graphics.

Course Outcomes		
Sr No	Code	CO statement
1	CO1	Understand the fundamental concepts of computer graphics, including the graphics pipeline, graphics APIs, and display technologies.
2	CO2	Develop and implement algorithms for drawing 2D primitives, polygon filling, and applying anti-aliasing techniques.
3	CO3	Utilize OpenGL for graphics programming, including creating animations and implementing basic 3D viewing and transformation operations.
4	CO4	Apply 2D and 3D geometric transformations, including affine transformations and coordinate system conversions, to manipulate graphical objects.

5	CO5	Demonstrate knowledge of viewing transformations, 3D projections, hidden surface removal techniques, and animation sequences for realistic graphics rendering.
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Unit No	Course Contents	Duration
I	Introduction to Computer Graphics: Overview of Computer Graphics, Computer Graphics Application and Software, Graphics Areas, Graphics Pipeline, Graphics API's, Numerical issues, Efficiency Display and Hardcopy Technologies, Display Technologies – Raster scan Display System	6 hours
II	Basic Raster Graphics: Algorithms for Drawing 2D primitives, aliasing and ant aliasing, Polygon filling methods: Scan Conversion Algorithms: Simple Ordered edge list, Edge Fill, Fence fill and Edge Flag Algorithm, Seed fill Algorithms: Simple and Scan Line Seed Fill Algorithm, Halftoning techniques.	7 hours
III	Graphics Programming using OpenGL: Why OpenGL, Features in OpenGL, OpenGL operations, Abstractions in OpenGL – GL, GLU & GLUT, 3D viewing pipeline, viewing matrix specifications, a few examples and demos of OpenGL programs, Animations in open GL.	7 hours
IV	2-Dgeometric transformations: Basic transformations, matrix representations, composite transformations, other transformations, transformations between coordinate systems, affine transformations, transformation functions, Raster methods for transformations. TwoDimensional viewing : viewing coordinates, Window-to viewport coordinate transformation, viewing functions, clipping : point, line, polygon, curve, text, exterior.	8 hours
V	Normalized Device Coordinates and Viewing Transformations:3D System Basics and 3D Transformations, 3D graphics projections, parallel, perspective, viewing transformations.3D graphics hidden surfaces and line removal, painter's algorithm, Z -buffers, Warnock's algorithm. Animations & Realism 10 Animation Graphics: Design of Animation sequences – animation function – raster animation – key frame systems – motion specification –morphing – tweening, Light Sources.	8 hours

Text Books				
Code	Title of the Book	Author Name /Designation / Organization	Publisher	Edition/ Publication Year
T1	Computer Graphics with OpenGL	Donald Hearn and Pauline Baker	Pearson	4th Edition, 2020
T2	Introduction to Computer Graphics and the Vulkan API	Kenwright	Cambridge Scholars Publishing	2021

Reference Books				
Code	Title of the Book	Author Name/Designation/ Organization	Publisher	Edition/ Publication Year
R1	Computer Graphics: Principles and Practice	John F. Hughes, Andries van Dam, Morgan McGuire, David F. Sklar, James D. Foley, Steven K. Feiner, and Kurt Akeley	Pearson	4th Edition, 2022
R2	Interactive Computer Graphics: A Top-Down Approach with WebGL	Edward Angel and Dave Shreiner	Pearson	8th Edition, 2022

Useful Links	
1	http://www.coursera.org/articles/computer-graphics
2	http://www.javatpoint.com/computer-graphics-tutorial